

25 SQL and Database Testing Q&A

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These questions cover the most practical SQL and database testing topics for QA engineers.

1. What is SQL?

Answer: SQL (Structured Query Language) is used to manage and query relational databases. It allows inserting, updating, deleting, and retrieving data.

2. What is database testing in QA?

Answer: Database testing ensures data integrity, correctness, and reliability by validating schemas, tables, triggers, procedures, and transactions.

3. What are the main types of database testing?

Answer:

- Functional Testing (CRUD operations, business rules)
- Data Integrity Testing
- Performance/Load Testing
- Security Testing

4. What is the difference between SQL and NoSQL databases?

Answer: SQL databases are relational and use structured schemas. NoSQL databases are non-relational, schema-less, and optimized for scalability.

5. How do you test data integrity?

Answer: By validating constraints (primary keys, foreign keys, unique, not null) and verifying data matches business rules.

6. What are primary keys and foreign keys?

Answer: A primary key uniquely identifies a row in a table. A foreign key establishes a relationship between two tables.

7. How do you test stored procedures?

Answer: Write test cases to verify inputs, outputs, and error handling. Check business logic correctness and performance.

8. What are indexes in SQL?

Answer: Indexes speed up query retrieval. QA engineers test performance with and without indexes to ensure efficiency.

9. What is normalization?

Answer: The process of organizing data into tables to reduce redundancy and improve integrity.

10. What is denormalization?

Answer: The process of combining tables to improve read performance, often used in data warehouses.

11. How do you test database performance?

Answer: By running load tests, checking execution plans, monitoring slow queries, and verifying response times.

12. What is ACID in databases?

Answer: ACID stands for Atomicity, Consistency, Isolation, Durability. It ensures reliable database transactions.

13. What is a transaction in SQL?

Answer: A transaction is a sequence of operations executed as a single logical unit of work.

14. How do you test triggers?

Answer: Validate that triggers fire under the right conditions and produce expected results without side effects.

15. What is data migration testing?

Answer: Testing that data moves correctly between systems with no loss, truncation, or corruption.

16. How do you validate data in ETL processes?

Answer: Compare source and target data, validate transformations, and check record counts.

17. What is SQL injection?

Answer: A security vulnerability where attackers manipulate SQL queries via user input. QA tests to ensure input sanitization.

18. How do you prevent SQL injection?

Answer: Use prepared statements, parameterized queries, and input validation.

19. What is the difference between INNER JOIN and OUTER JOIN?

Answer:

- INNER JOIN returns matching rows from both tables.
- OUTER JOIN returns matching rows plus non-matching rows from one or both tables.

20. How do you test database backups?

Answer: By restoring backups and validating that data and schema match the original.

21. What are common SQL constraints to test?

Answer: Primary Key, Foreign Key, Unique, Check, Not Null, Default.

22. What is a deadlock in SQL?

Answer: A situation where two or more transactions block each other indefinitely. QA checks concurrency scenarios.

23. How do you identify duplicate records in SQL?

Answer: Using GROUP BY with HAVING COUNT(*) > 1 or ROW_NUMBER() window functions.

24. What are database mock/stub strategies in QA?

Answer: Using mock data or stubs to simulate database interactions for isolated testing.

25. What are best practices in database testing?

Answer:

- Use production-like data sets
- Validate data integrity and business rules
- Automate repetitive SQL queries
- Ensure performance under load